ModelIt!

Teacher's Guide

Earth Systems Collide: Volcanoes, Climate, and Life

How a Volcanic Eruption Can Change the Whole Planet

Quick Reference

Grade Level	Grade 05L2
NGSS Standard	5-ESS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-ESS2-1

Develop a model using an example to describe ways the geosphere, biosphere, hydrosphere, and/or atmosphere interact.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing interactions between the geosphere, atmosphere, biosphere, and hydrosphere.
Disciplinary Core Idea	**ESS2.A Earth Materials and Systems** — Students model how volcanic activity affects atmospheric CO2 and global temperature.
Crosscutting Concept	**Systems and System Models** — Students model Earth as a system with feedback loops that promote stability.

CAST Testing Alignment

MULTIPLE CHOICE:

After a massive volcanic eruption, atmospheric CO2 rises sharply. Which of the following is a negative feedback that helps reduce CO2?

CONSTRUCTED RESPONSE:

Explain how a volcanic eruption in Asia could affect the climate in North America, using evidence from your model.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Earth is a system of interacting spheres — the geosphere (rock), atmosphere (air), hydrosphere (water), and biosphere (life). Volcanic eruptions are a dramatic example of how one sphere's activity can ripple through all the others.

Volcanoes and CO2

When volcanoes erupt, they release large amounts of CO2 and other gases into the atmosphere. Over geologic time, volcanic CO2 has been a major driver of climate change.

The Greenhouse Effect

CO2 is a greenhouse gas that traps heat in the atmosphere. More CO2 means more heat is retained, raising global temperatures. This is the greenhouse effect.

Negative Feedback: Vegetation

When CO2 and temperature rise, plants grow faster because they have more CO2 for photosynthesis. This increased growth absorbs some of the extra CO2, partially counteracting the warming.

Negative Feedback: Ocean Absorption

Oceans absorb about 30% of atmospheric CO2. As CO2 levels rise, oceans absorb more. However, this also makes oceans more acidic, which harms marine life.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Volcanic Activity, Atmospheric CO2, Global Temperature, Ocean Absorption, and Vegetation Growth as the five system components.

E **Establish Relationships**

Students determine that Volcanic Activity increases CO2 (+), CO2 increases Temperature (+), Temperature increases Vegetation (+), Vegetation decreases CO2 (-), and Ocean Absorption decreases CO2 (-).

V **Visualize & Model**

Students build the Earth systems model showing how volcanic CO2 drives warming while vegetation and oceans create negative feedback loops.

E **Evaluate Outcomes**

Students run the Massive Eruption scenario and observe CO2 and temperature spiking, then the Self-Regulation scenario to see feedback loops in action.

R **Revise & Extend**

Students add Volcanic Ash (cooling effect), Ice Cap Melting, or Human CO2 Emissions to explore more complex Earth system interactions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"A volcano erupts in Asia. Could it change the weather in your town?"*

Watch For

Show eruption image and ask: What comes out of a volcano besides lava?

Slide 2: Learning Objectives

2 min

Say: *"Today we model how Earth's four spheres interact during a volcanic eruption."*

Watch For

Students list the four spheres: geosphere, atmosphere, biosphere, hydrosphere.

Slide 3: Big Question

2 min

Say: *"Volcanoes don't just affect the nearby area. Their impact is GLOBAL."*

Watch For

Quick pair-share: How could volcanic gas travel around the world?

Slide 4: LEVER Framework

2 min

Say: *"Our model includes a new concept: negative feedback loops."*

Watch For

Define negative feedback and preview how it appears in the model.

Slide 5: Activity 1: Components

8 min

Say: *"Volcanic Activity is our only external input. Everything else responds in a chain."*

Watch For

Students sort components and predict which sphere each belongs to.

Slide 6: Activity 2: Connections

8 min

Say: *"Look at the arrows from Vegetation and Ocean back to CO_2. Those are negative feedback loops — Earth fighting back."*

Watch For

Students draw all five relationships and identify the two feedback loops.

Slide 7: Activity 3: Simulation

10 min

Say: *"Massive eruption! Watch CO_2 and temperature spike — then watch Earth respond."*

Watch For

Students run Massive Eruption and Self-Regulation scenarios.

Slide 8: Discoveries

5 min

Say: *"Earth has built-in stabilizers. But can they handle everything we throw at them?"*

Watch For

Class discussion: How is volcanic CO_2 different from human CO_2 emissions?

Slide 9: STEM Challenge

10 min

Say: *"If Yellowstone erupted, how would we track the climate changes?"*

Watch For

Groups design monitoring systems; present to class.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Volcanic Activity
- **INTERNAL:** Atmospheric CO2, Global Temperature, Ocean Absorption, Vegetation Growth

Relationships

- **Volcanic Activity to Atmospheric CO2:** POSITIVE (+) — Volcanic eruptions release large amounts of CO2 into the atmosphere.
- **Atmospheric CO2 to Global Temperature:** POSITIVE (+) — More CO2 traps more heat, increasing Earth's average temperature.
- **Global Temperature to Vegetation Growth:** POSITIVE (+) — Warmer temperatures and more CO2 stimulate plant growth (to a point).
- **Vegetation Growth to Atmospheric CO2:** NEGATIVE (-) — More vegetation absorbs more CO2 through photosynthesis, reducing atmospheric levels.
- **Ocean Absorption to Atmospheric CO2:** NEGATIVE (-) — Oceans absorb CO2 from the atmosphere, reducing atmospheric levels.

Key Discoveries

- Volcanic eruptions add CO2 to the atmosphere, which raises global temperature
- Warmer temperatures and more CO2 cause plants to grow faster, absorbing some CO2 (negative feedback)
- Oceans also absorb CO2, helping to regulate the atmosphere (another negative feedback)
- Earth has natural self-regulation systems, but they can be overwhelmed by extreme events

Multiple Choice Post-Assessment Answer Key

1. In the Earth systems model, what happens when Volcanic Activity is set to 90%?

Correct: B) Atmospheric CO2 spikes, which causes Global Temperature to rise

Correct! Volcanic eruptions release massive amounts of CO2 into the atmosphere. CO2 is a greenhouse gas that traps heat, so more CO2 leads to higher global temperatures. The model shows this positive chain: Volcanic Activity increases CO2, and CO2 increases Temperature.

2. What is a 'negative feedback loop' in the Earth systems model?

Correct: B) A response that counteracts a change, pushing the system back toward balance

That's right! A negative feedback loop works against a change. When CO2 rises, vegetation grows faster and absorbs more CO2, partially reducing the increase. The response pushes back against the original change, helping stabilize the system.

3. How does Vegetation Growth create a negative feedback loop with Atmospheric CO2?

Correct: B) When CO2 rises, plants grow faster and absorb more CO2 through photosynthesis, pulling CO2 back down

Correct! Rising CO2 stimulates plant growth because CO2 is a raw material for photosynthesis. As plants grow faster, they absorb more CO2, reducing atmospheric levels. This loop counteracts the original CO2 increase, which is why it is called negative feedback.

4. In the Self-Regulation scenario (Volcanic Activity at 60%), what does the model demonstrate about Earth's climate?

Correct: C) The system reaches a new, higher equilibrium because feedback loops partially counteract the CO2 input

Exactly! At moderate volcanic activity, CO2 and temperature rise, but vegetation and ocean absorption increase enough to partially offset the input. The system stabilizes at a new, slightly higher level rather than spiraling out of control. This shows Earth's self-regulating ability.

5. How can a volcanic eruption in Asia affect the climate in North America?

Correct: B) Volcanic gases and CO_2 spread through the atmosphere globally, affecting temperature and weather patterns worldwide

Correct! Volcanic gases and CO_2 enter the atmosphere and can circle the entire globe within weeks. The greenhouse effect from added CO_2 warms the whole planet, not just the area near the volcano. This is how Earth's spheres interact globally.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Volcanoes only affect the area nearby"	Volcanic gases and ash can circle the entire globe within weeks. Major eruptions have changed global temperatures for years.
"Earth's temperature is always the same"	Earth's temperature has changed dramatically over geologic time, driven by volcanic activity, solar changes, and CO2 levels.
"Plants can solve climate change on their own"	While plants absorb CO2, there are limits. Deforestation is removing plants, and human CO2 emissions are far greater than what current vegetation can absorb.